FinAcumen: Financial Multimodal Reasoning via Self-Evolving Experience Memory Harness

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HarnessRSI 68 ★★

evidence: retained

Paper

Code

1 What changes, what is kept

BUILD · offline, from train/validation items

1. **Sample K = 8 trajectories**
   per item, same frozen model and tools
2. **Score each against the gold answer**
   correct = success, wrong = failure
3. **Summary agent writes experience**
   Findings from wins, Cautions from failures
4. **Add entry to the bank, then freeze**
   question + gold answer + Findings + Cautions; about 2,400 entries; read-only at test

ANSWER · each test question

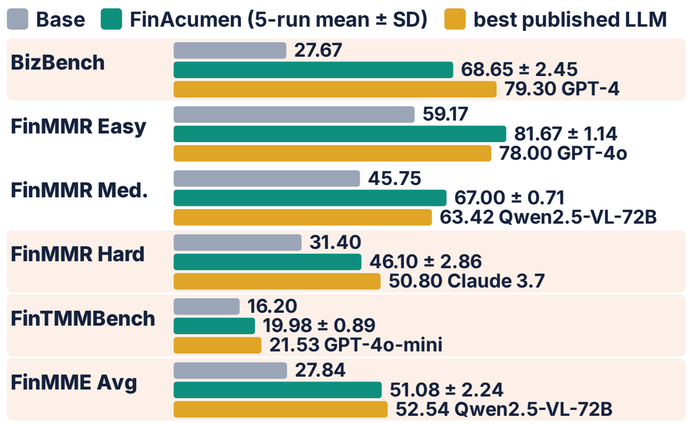
5. **Retrieve**
   embed question + context; keep entries with cosine ≥ τ = 0.65; de-duplicate; top k_max = 5

- **no entry ≥ τ**: Fallback: solver runs with the tools only, no memory
- **≥ 1 entry**: Annotator rewrites them into directives for this question

6. **Solve**
   ReAct loop ≤ 16 steps with 4 fixed tools; answer gate checks source, units, format

Weights and tools never change: the only thing built is a text bank

2 Headline: +3.78‡ to +40.98‡ over Base



Tables 1, 9: frozen Qwen3-VL-8B, 5-run mean. Shaded: below the best published model (model variants per Table 8, copied not re-run).

3 Controls: is the gain from memory?

accuracy	Base	+ tools	+ memory	tools Δ	mem. Δ
FinMMR Easy (T2)	59.17	74.08	**81.67**	+14.91	**+7.59**
BizBench (Fig. 2)	27.67	55.15	**68.65**	+27.48‡	**+13.5**
BizBench 235B (T12)	57.45	69.30	**75.10**	+11.85	**+5.80**

Only 3 settings are split; tools give about 2× the memory gain. No SD.

same tools + data	Reflexion	ExpeL	FinAcumen
FinMMR Easy	77.17	77.17	**81.67**
FinMMR Medium	63.58	60.58	**67.00**
FinTMMBench	14.52	16.73	**19.98**
FinMME	42.97	45.77	**51.08**

App. I, Table 10: 4 settings, single numbers, no SD, tuning not described.

Table 4 (600-q subsample, τ 0.60): experience +3.50 / +9.00‡ vs tools + annotation +2.25.

4 Where the gain is absent, small or untested

- **Below the best published model in 4 of 6 columns:** BizBench 68.65 vs 79.30, FinMMR Hard 46.10 vs 50.80, FinTMMBench 19.98 vs 21.53, FinMME 51.08 vs 52.54. (text lists three)
- **FinTMMBench: smallest gain** (16.20 to 19.98); Reflexion 14.52 and ExpeL 16.73 sit near Base.
- **FinMME:** memory hit rate 0.48, mean 1.15 entries per query (1200-entry snapshot, Table 5); no tools/memory split shown.
- **Selectivity not isolated:** no accuracy ablation of the τ gate / fallback; Findings and Cautions never ablated separately.
- **Cost:** tokens per question 20,043.51 to 42,523.13 with the bank; build CNY 142.97, 78.12 h; no matched-compute baseline.
- **Inconsistencies:** bank about 2,400 vs about 1800 entries; 11,379 test queries (App. E) vs Table 6 sizes summing to 8,756‡.

Takeaway

Offline experience adds to fixed tools on a frozen 8B model. Where the paper splits them (3 settings) tools give about 2× the memory gain (+7.59, +13.5), though its Table 4 subsample makes experience the larger step. Ahead of Reflexion / ExpeL on four settings (single numbers, no SD). Four columns still trail the best published model; the τ gate is untested against always injecting.

Our read · HarnessRSI

Score 68 · ★★ · evidence level: retained

Reports: held-out bank source, 5 runs with CIs, tools vs memory (3 configs), Reflexion/ExpeL, cost, other backbones.

Does not report: matched compute, overlap check bank vs test, τ-gate ablation, cross-benchmark transfer.